CentralFormer: Centralized Spectral–Spatial Transformer for Hyperspectral Image Classification With Adaptive Relevance Estimation and Circular Pooling

Ningyang Li, Zhaohui Wang, *Member, IEEE*, Faouzi Alaya Cheikh, *Senior Member, IEEE*, and Lei Wang

***Abstract*— Classification of hyperspectral image (HSI) is a hotspot in the field of remote sensing. Recent deep learning (DL)-based approaches, especially for the transformer architectures, have been investigated to extract the deep spectral–spatial features. However, the ability of these approaches to efficiently represent the crucial attention patterns and distinguishing features suffers from the neglect of the relevant areas, including the center pixel and high computational complexity; thereby, their classification performances still need to be improved. This article proposes a centralized spectral–spatial transformer (CentralFormer), which contains the central encoder, the adaptive relevance estimation (ARE) module, and the cross-encoder relevance fusion (CERF) module, for HSI classification. To recognize the relevant areas, the ARE modules access both spectral and spatial associations between the center pixel and its neighborhoods flexibly. By focusing on these areas and emphasizing them during attention inference, the central encoders can extract the key attention modes and discriminating features effectively. Moreover, the CERF modules are deployed to prevent the reliability of the relevance map from being harmed by the feature deviation between encoders. To handle the high computational occupancy, a novel circular pooling (CP) strategy reduces the circles and bands of features. Unlike regular pooling methods, it can well improve the relevant characteristics for subsequent encoders. By integrating these techniques, the CentralFormer model can represent the discriminating spectral–spatial features efficiently for HSI classification. Experimental results on four classic HSI datasets reveal that the proposed CentralFormer model outperforms the state-of-the-arts in terms of both classification accuracy and computational efficiency. The source code is available at https://github.com/ningyang-li/CentralFormer.**

***Index Terms*— Attention mechanism, center pixel, circular pooling (CP), hyperspectral image (HSI) classification, relevant area, transformer.**

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I. Introduction

UNPRECEDENTED advances of imaging devices and propulsion technology provide Earth observation with a firm foundation. Hyperspectral imaging, one of the most popular remote sensing means, can portray both continuous spectral signatures and complicated spatial organizations of land covers [1], [2]. Benefits from this rich information, hyperspectral image (HSI) classification has been the primary tool for smart management in the fields of agriculture [3], urban planning [4], fishery [5], and so on.

Over the past more than 20 years, a variety of algorithms have been proposed to reach better classification performances. Some early traditional classification algorithms, such as support vector machine (SVM) [6], k-nearest neighbor (KNN) [7], and decision tree [8], were employed to recognize raw spectra and bands. However, it is generally difficult for those methods to leverage the important information, including corners, key points, and textures. Hence, scale-invariant feature transform [9], histogram of gradient [10], morphological profile [11], and so on were utilized to extract the complex handcrafted features. Although these solutions have improved classification performances to some extent, the effectiveness of these shallow handcrafted features tends to be reduced because of the redundant bands and spectral variability [12]. Deep feature representation has been an urgent requirement for elevating the robustness of classification results.

In recent years, extraordinary progress has been made in deep learning (DL) with the upgrades of graphics processing hardware. Various neural networks were proposed to handle image classification [13], target detection [14], semantic segmentation [15], and so on. In the field of HSI classification, researchers have devoted significant effort in improving the effectiveness of spectral–spatial features. The common models used to extract spectral features, such as multilayer perceptron [16], deep belief network [17], autoencoder [18], recurrent neural network [19], and long short-term memory

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network [20], generally deal with each spectrum as a sequence and receive a spectral response with an exclusive neuron. Both spectral characteristics and the dependence between bands can be well represented by these models. Nevertheless, their classification accuracy needs to be increased yet on account of the absence of spatial features.

The convolutional neural network (CNN) [13], which excels at sensing local structures, is the classic architecture to extract spatial features of HSIs [21]. Its convolutional kernels, also named filters, play a crucial role in feature extraction. On the one hand, the flexible size of convolutional kernels enables the spatial features in different scales, which can be described in detail for classification [22]. On the other hand, CNNs are able to express both spectral and spatial features with the dimensional transformation of kernels [23]. As a result, convolution has been integrated with many popular architectures, including a generative model [26], a graph network [25], and a long short-term memory network [24], to gather more useful local details. By stacking multiple convolutional layers [27], deep CNNs are good at representing more abstract spectral–spatial features in different patterns [28]. However, the phenomenon of network degradation will occur, while networks become deeper. To resolve this problem, residual learning [29] adopted the shortcut connections between output and previous features, which allows CNNs to develop in any depth. In addition, a densely connected network [30] and a capsule network [31] employed the techniques of feature accumulation and multidimension neurons to improve the density and hierarchy of features, separately. With these tricks, the spectral–spatial models [32], [33], [34], [35], [36], especially for the 3-D CNN-based models, have been the dominant architectures to obtain the deep and complicated spectral–spatial features for HSI classification.

Considering that an HSI cube contains hundreds of bands and many pixels, those spectral–spatial features from each band and position actually do not exert the same influence on classification. In recent years, a feature selection approach in the field of neural machine translation named attention mechanism [37] was introduced to highlight the salient bands and useful pixels in samples. The most popular attention modules so far for HSI classification are the squeeze-and-excitation (SE) module [38], the convolutional block attention module (CBAM) [39], and the self-attention (SA) module [40]. The SE module tends to infer the key bands by transforming the global spectral semantic information [41]. Besides the salient bands, CBAM can also explore the crucial pixels by gathering spatial intensity information [42]. Unlike the former two, the SA module takes into account the correlations between pixels to acquire that important information [43]. However, the SA module generally spends a considerable long time on attention inference due to its massive matrix operations. These attention modules are often integrated into different parts of backbones to recalibrate the features in different stages [44].

However, the three kinds of modules can only capture the relatively single attention in the global view. Hence, transformer architectures [45], [46], [47], the assemblies of many SA modules, were built to generate more patterns of attention maps in multiple subspaces. During these processes, transformer architectures generally received the spectral–spatial embeddings [48] and leveraged the global spectral and spatial correlations to deduce the vital bands and pixels [49], [50]. To enhance the feature representation further, transformer architectures and CNNs were combined to capture both global dependence and local specifics [51], [52], [53]. These models tend to extract the common spectral–spatial features, which generally not consider the characteristics of the relevant areas, especially the center pixel. Recent studies [54], [55] attempted to exert the influence of the center pixel on the input embeddings, but the center pixel was absent yet during attention deduction. Consequently, the classification performances of these methods need to be improved.

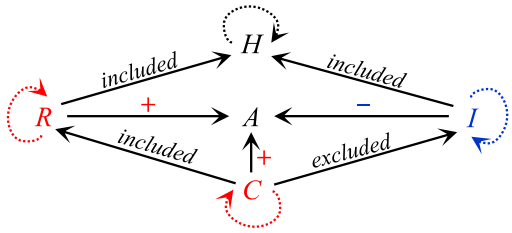


Fig. 1. Relationships among HSI cube (H), the center pixel (C), the relevant pixels (R), the interfering pixels (I), and attention (A), where "+" and "−" denote promotion and suppression effects, respectively.

In summary, there are still two notable deficiencies for current transformer architectures. One is that they cannot recognize and concentrate on the relevant pixels closely connected to the center pixel during both attention generation and feature extraction. As shown in Fig. 1, due to the special sampling mechanism, an HSI cube contains generally the relevant pixels that own the same class as the center pixel and other interfering pixels. The relevant pixels are beneficial to explore the effective attention modes, thereby discriminating features [42], [56], [57]. However, most transformer architectures are weak in distinguishing this discrepancy and can hardly focus on the dependencies between the relevant pixels and others during attention inference, which weakens the effectiveness of attention and features and limits the classification performances. The other is the heavy computational complexity of transformer architectures [48], [50], [51], [52], [54], [55]. Pooling, including max-pooling (MP) and average pooling (AP), is a popular way to reduce the time and memory consumption for attention deduction [49], [53]. However, the characteristics of the relevant areas, especially for the center pixel, will be destroyed in these regular pooling patterns, which deviates from the reasoning of relevant attention and lowers the classification accuracy.

To handle both deficiencies, this article proposes a centralized spectral–spatial transformer (CentralFormer) to infer the attention related to the relevant areas and extract the discriminating features efficiently for HSI classification. It is mainly composed of the central encoder, the adaptive relevance estimation (ARE) module, and the cross-encoder relevance fusion (CERF) module. The ARE modules first recognize the relevance areas based on both spectral and spatial correlations between the center pixel and others. Guided by the relevance

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areas, the central encoders can then explore more crucial attention patterns and relevant spectral–spatial features. The CERF modules are also deployed between central encoders to consider different levels of relevance maps, which consolidates the reliability of the relevance maps when the model becomes deeper. Moreover, to improve the computational complexity and classification accuracy, a novel circular pooling (CP) method is developed to reduce the dimensions of features while maintaining the characteristics of relevant areas, especially the center pixel. Therefore, the proposed model can extract the discriminating spectral–spatial features efficiently for HSI classification. Experimental results on four classic HSI datasets confirm the structural effectiveness and outstanding classification performances of the CentralFormer model.

The main contributions of this article are given as follows.

1) A central encoder, which contains a spectral–spatial local representation (SSLR) module, a centralized multihead attention (CMHA) module, a feature fusion part (FFP), and a CP layer, is designed to deduce the relevant attention modes and acquire more effective spectral–spatial features. The SSLR module aims to represent the local spectral dependence and spatial structures. The CMHA module processes the relevant areas as the key set and emphasizes the dot-product similarity of them via 1-D Gaussian function, thereby providing the important attention modes and global features. After the feature fusion, a novel CP strategy, which regards an HSI cube as the center pixel and several circles, reduces the circles and bands of features. Unlike regular pooling ways, it protects the characteristics of relevant areas saliently, which is the core of improving computational complexity and subsequent features.
2) To promote the involvement of relevant areas in attention deduction and feature extraction, an ARE module easy to optimize is developed to discover the pixels relevant to the center pixel. Both spectral angle distances and discrete Gaussian distances between the center pixel and its neighborhoods are integrated adaptively during this process.
3) A CERF module is built to relieve the negative influence of network depth on the reliability of relevance maps. It can fuse the current relevance map and the previous one adaptively to take into full account their relative importance. Before fusing, the CP layer is also utilized to adjust the size of the previous relevance map and maintain its relevant peculiarity.
4) An end-to-end CentralFormer model is proposed for HSI classification. Compared with other methods, especially for the transformer architectures, it excels at inferring the critical attention modes in connection with the relevant areas and extracting the discriminating spectral–spatial features efficiently.

The remainder of this article is arranged as follows. Section II describes the CentralFormer model in detail. Experimental results and analyses are presented in Section III. Finally, Section IV summarizes this article.

II. Methodology

A. Overview of the Proposed Model

As shown in Fig. 2, the CentralFormer model is mainly composed of the central encoders, ARE modules, and CERF modules. Given a raw HSI cube $\boldsymbol{X} \in \mathbb{R}^{1\times w_1\times w_1\times b_1}$, the information between adjacent bands is first collected into a spectral embedding $\boldsymbol{F}_1 \in \mathbb{R}^{1\times w_1\times w_1\times b_1}$ with a $1 \times 1 \times 3$ convolutional kernel, where "1" indicates an additional channel dimension and "w_i" and "b_i" denote the width and number of bands of the input features of the ith central encoder, separately.

In the first central encoder, the local features $\boldsymbol{L}_1 \in \mathbb{R}^{h_1\times w_1\times w_1\times b_1}$ are obtained by the SSLR module from the channel-increased input $\boldsymbol{F}_1^{ci} \in \mathbb{R}^{h_1\times w_1\times w_1\times b_1}$, where "$h_i$" denotes the number of attention heads (channels). To take the positive information into account, the ARE module is exploited in advance to generate the relevance map $\boldsymbol{M}_1 \in \mathbb{R}^{1\times w_1\times w_1\times 1}$ based on both spectral and spatial correlations between the center pixel and its neighborhoods. With the assistance of the relevance map $\boldsymbol{M}_1$, the CMHA module then explores the key attention modes and extracts the discriminating global features $\boldsymbol{A}_1 \in \mathbb{R}^{h_1\times w_1\times w_1\times b_1}$ from previous local features $\boldsymbol{L}_1$. Next, both local and global features are fused to derive the comprehensive representation $\boldsymbol{F}_1' \in \mathbb{R}^{h_1\times w_1\times w_1\times b_1}$. To promote the optimization of the whole encoder, the skip connection is also used to obtain the deep residual features $\boldsymbol{F}_1'' \in \mathbb{R}^{h_1\times w_1\times w_1\times b_1}$. Finally, a CP layer reduces the circles and bands of features but well preserves the characteristics of relevant areas in resultant pooled features $\boldsymbol{F}_2 \in \mathbb{R}^{h_1\times w_2\times w_2\times b_2}$.

Before the second central encoder, the relevance map $\boldsymbol{M}_2' \in \mathbb{R}^{1\times w_2\times w_2\times 1}$ in corresponding features $\boldsymbol{F}_2$ is reevaluated by the ARE module. It is fused with previous relevance map $\boldsymbol{M}_1$ adaptively by the CERF module to improve the precision of the final relevance $\boldsymbol{M}_2 \in \mathbb{R}^{1\times w_2\times w_2\times 1}$. Following the similar processes, the latter two central encoders extract the discriminating spectral–spatial features $\boldsymbol{F}_3 \in \mathbb{R}^{h_2\times w_3\times w_3\times b_3}$ and $\boldsymbol{F}_4 \in \mathbb{R}^{h_3\times w_4\times w_4\times b_4}$, separately. $\boldsymbol{F}_4$ will be sent to a fully connected layer to acquire the softmax probability vector $\boldsymbol{y} \in \mathbb{R}^{1\times N}$, where N denotes the number of classes.

B. Central Encoder

As one of the core components of the proposed model, the central encoder processes spectral embedding or features and outputs the discriminating spectral–spatial features. In this section, the structures of the central encoder, including the SSLR module, the CMHA module, the FFP, and the CP layer, are detailed.

1) SSLR Module: The SSLR module is built to acquire the spectral dependence that existed in adjacent bands and local spatial structures. The architecture of the SSLR module is shown in Fig. 3. It is composed of a 3-D pixelwise convolution (PWConv3D) layer, a 3-D bandwise convolution (BWConv3D) layer, and a spectral–spatial aggregation part.

Taking the ith central encoder as an example, the short-term spectral dependence $\boldsymbol{L}_i^{\boldsymbol{e}} \in \mathbb{R}^{k_{\text{exp}}\times w_i\times w_i\times b_i}$ hid in adjacent bands is discovered in the PWConv3D layer with $1 \times 1 \times 7$ convolutional kernels. The number of kernels k_{exp} in this layer is higher than the number of channels of input $\boldsymbol{F}_i^{ci} \in$

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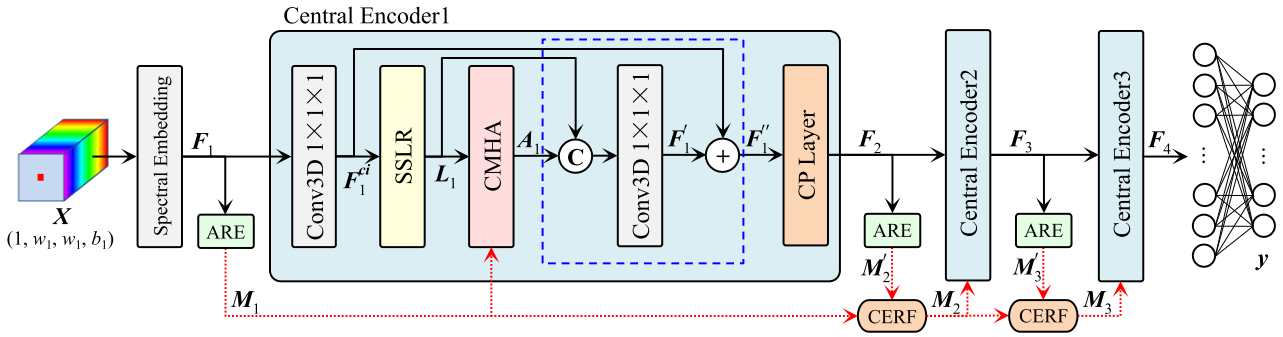


Fig. 2. Overview of the CentralFormer model. It contains three central encoders, where blue dotted box, "C," and "+" indicate FFP, concatenation, and elementwise addition, respectively. The ARE and CERF modules are designed to capture the relevant areas in different levels of features.

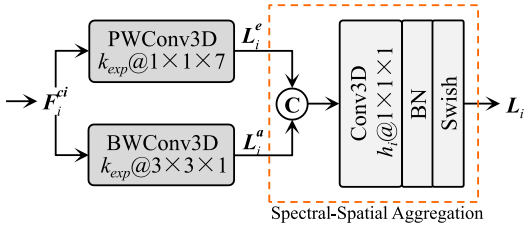


Fig. 3. Architecture of the SSLR module.

$\mathbb{R}^{h_i \times w_i \times w_i \times b_i}$. The local spatial structures $\boldsymbol{L}_i^a \in \mathbb{R}^{k_{\exp} \times w_i \times w_i \times b_i}$ are obtained efficiently by the BWConv3D layer, which has $k_{\exp}$ kernels with the size of $3 \times 3 \times 1$. Compared with regular convolution, PWConv3D and BWConv3D layers can explore more patterns of spectral and spatial features with fewer parameters [58], [59]. The separated convolutional kernels also reduce the interference between spectral and spatial dimensions.

Both kinds of local representation are integrated with the $1 \times 1 \times 1$ convolutional layer of spectral–spatial aggregation part (orange dotted box). After regularizing data distribution, the Swish [60] activation function, which is easier to be optimized compared with other functions, is adopted to reinforce the nonlinear attributes of local features $\boldsymbol{L}_i \in \mathbb{R}^{h_i \times w_i \times w_i \times b_i}$.

2) CMHA Module: As the core ideology of transformer architectures, multihead attention is good at capturing the global correlations between all pixels. However, because of the special sampling strategy of the HSI cube, the effective information intensively connected to the center pixel generally is not valued with the original multihead attention. To handle this deficiency, the CMHA module is proposed to pay more attention to the associations between the relevant pixels and others so that the discrimination of global features is enhanced.

The architecture of the CMHA module is shown in Fig. 4. It extracts the global features $\boldsymbol{A}_i$ from the local features $\boldsymbol{L}_i$ of the ith central encoder. Unlike common transformer architectures, the key set of the CMHA module is improved to the relevance set $\boldsymbol{R}_i \in \mathbb{R}^{h_i \times w_i^2 \times b_i}$, which is acquired with the guidance of relevance map $\boldsymbol{M}_i$ (red dotted arrows), where h_i denotes the number of attention heads of the ith central encoder. This is to connect the attention in different subspaces with the relevant pixels, thereby ensuring the discrimination of global features. After this process, the query set $\boldsymbol{Q}_i \in \mathbb{R}^{h_i \times w_i^2 \times b_i}$, the relevance set $\boldsymbol{R}_i$, and the value set $\boldsymbol{V}_i \in \mathbb{R}^{h_i \times w_i^2 \times b_i}$ are obtained with three groups of linear layers

$$\boldsymbol{Q}_i = \text{Linear_}Q(\boldsymbol{L}_i) \tag{1}$$

$$\boldsymbol{R}_i = \text{Linear_}R(\boldsymbol{L}_i \circledcirc \boldsymbol{M}_i) \tag{2}$$

$$\boldsymbol{V}_i = \text{Linear_}V(\boldsymbol{L}_i). \tag{3}$$

There are h_i centralized dot-product attention (CDPA) blocks in the CMHA module to capture the attention from multiple spaces. Taking the first CDPA block as an example, the matrix multiplication is first conducted to measure the dot-product similarity $\boldsymbol{U}_i^1 \in \mathbb{R}^{1 \times w_i^2 \times w_i^2}$ between the relevant areas and all pixels as follows:

$$\boldsymbol{U}_i^1 = \boldsymbol{Q}_i^1 \otimes \boldsymbol{R}_i^{1T} \tag{4}$$

where $\boldsymbol{Q}_i^1 \in \mathbb{R}^{1 \times w_i^2 \times b_i}$, $\boldsymbol{R}_i^1 \in \mathbb{R}^{1 \times w_i^2 \times b_i}$, and $\boldsymbol{V}_i^1 \in \mathbb{R}^{1 \times w_i^2 \times b_i}$ denote the query, relevance, and value sets of the first attention head of the ith central encoder, respectively.

Next, to ensure that attention inference considers more relevant pixels, the priorities of relevant pixels including the center pixel are enhanced in the centralization part. Specifically, the dot-product similarity between each pixel and itself in the diagonal of $\boldsymbol{U}_i^1$ (red dotted boxes) is extracted and expanded (EE) to each row of the diagonal similarity $\boldsymbol{D}_i^1 \in \mathbb{R}^{1 \times w_i^2 \times w_i^2}$. To increase the importance of each relevant pixel, the 1-D Gaussian function $\mathcal{G}_1$ is adopted to transform the difference map $\boldsymbol{S}_i^1 \in \mathbb{R}^{1 \times w_i^2 \times w_i^2}$ between dot-product similarity $\boldsymbol{U}_i^1$ and diagonal similarity $\boldsymbol{D}_i^1$ to the centralized dot-product similarity $\boldsymbol{C}_i^1 \in \mathbb{R}^{1 \times w_i^2 \times w_i^2}$

$$\boldsymbol{C}_i^1 = e^{-\boldsymbol{S}_i^{1^2}} = e^{-(\boldsymbol{U}_i^1 \ominus \mathbf{D}_i^1)^2}. \tag{5}$$

As shown in Fig. 5, the closer to zero the differences of $\boldsymbol{S}_i^1$, the higher the importances of $\boldsymbol{C}_i^1$. The smooth gradient of the 1-D Gaussian function assures the stability of optimization.

Then, the scaling operation and the softmax activation function are employed to obtain the attention map $\boldsymbol{W}_i^1 \in \mathbb{R}^{1 \times w_i^2 \times w_i^2}$

$$\boldsymbol{W}_i^1 = \text{softmax}\left(\frac{\boldsymbol{C}_i^1}{\sqrt{b_i}}\right) \tag{6}$$

where b_i indicates the number of bands of local features $\boldsymbol{L}_i$ in the ith central encoder.

After refining the value sets $\boldsymbol{V}_i = (\boldsymbol{V}_i^1, \boldsymbol{V}_i^2, \ldots, \boldsymbol{V}_i^{h_i})$ with corresponding attention maps $\boldsymbol{W}_i = (\boldsymbol{W}_i^1, \boldsymbol{W}_i^2, \ldots, \boldsymbol{W}_i^{h_i})$, the

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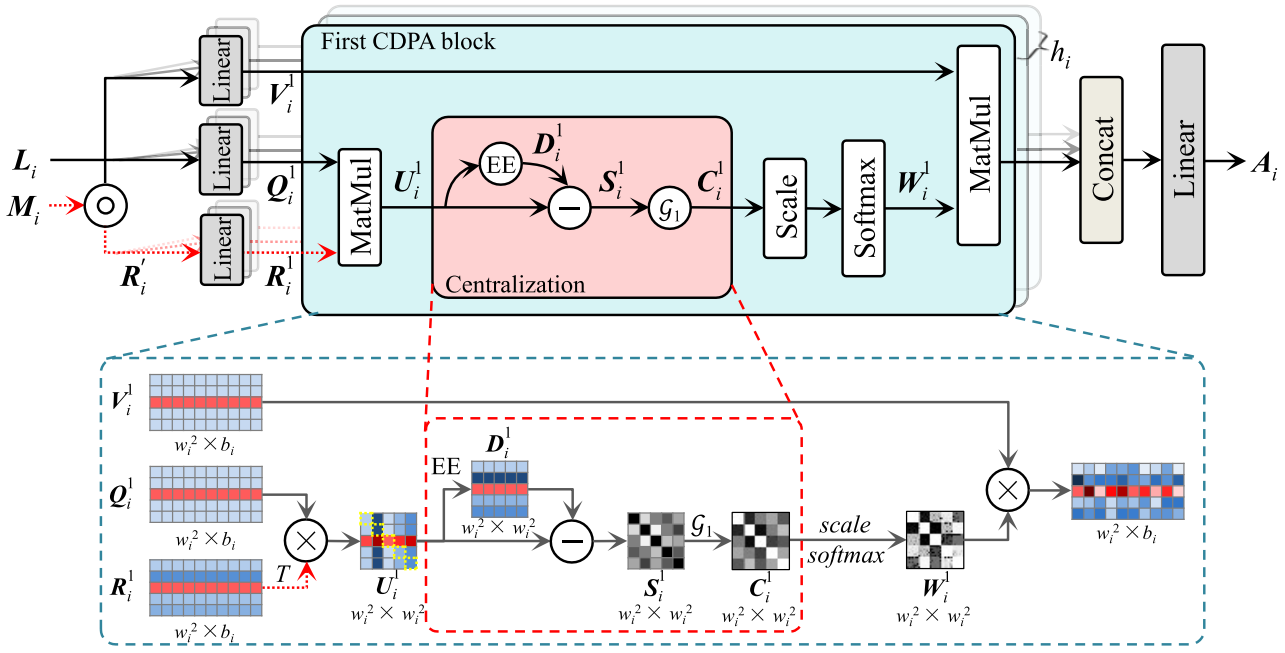


Fig. 4. Architecture of the CMHA module. The transformation of feature maps of the first CDPA block is shown in the lower part, where "⊙," "⊗," "T," and "⊖" mean the Hadamard product, matrix multiplication, transposition, and elementwise subtraction, respectively.

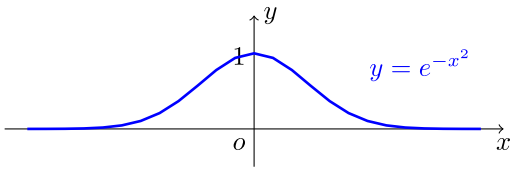


Fig. 5. One-dimensional Gaussian function $\mathcal{G}_1$.

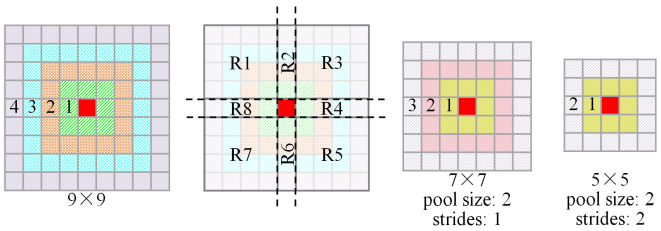


Fig. 6. Process of CP in spatial dimension. All pixels in feature maps are divided into nine regions. The region "R0" is actually the center pixel. Average operation is implemented in each region to reduce the circles.

global features of relevant areas are produced by another linear layer

$$\boldsymbol{A}_i = \text{Linear}(\boldsymbol{W}_i \otimes \boldsymbol{V}_i) \tag{7}$$

where $j = 1, 2, \ldots, h_i$ indicates the index of the CDPA block.

3) FFP: To enrich the diversity of features and improve classification performance, the global features $\boldsymbol{A}_i$ and local features $\boldsymbol{L}_i$ instead of the input $\boldsymbol{F}_i^{ci}$ are fused in the ith central encoder. This is due to that local features have a smaller feature gap with global features compared with input [59]. As shown in Fig. 2, a $1 \times 1 \times 1$ convolutional layer is used to gain the comprehensive representation $\boldsymbol{F}_i' \in \mathbb{R}^{h_i \times w_i \times w_i \times b_i}$. Using the convolutional layer instead of a single concatenation maintains the number of channels and also reduces the parameters of subsequent layers indirectly.

After fusion, the skip connection between comprehensive representation $\boldsymbol{F}_i'$ and input $\boldsymbol{F}_i^{ci}$ is built to construct the deep residual representation $\boldsymbol{F}_i'' \in \mathbb{R}^{h_i \times w_i \times w_i \times b_i}$, which can promote the optimization of deeper layers.

4) CP Layer: Due to the special structure of samples, the features extracted after the aforementioned processes generally contain redundant information both in spatial and spectral dimensions. Pooling is a popular strategy to reduce feature dimensions. However, the common pooling patterns, such as AP and MP, consider hardly the unique sampling mechanism of the HSI cube. The characteristics of features in the relevant areas, especially for the center pixel, will be weakened or confused with neighboring pixels to different degrees in those patterns, which hinders the expressiveness of subsequent features. To resolve this problem, a novel CP layer is designed in this work to maintain the effectiveness of relevant areas, including the center pixel, when reducing feature dimensions.

Unlike regular pooling methods, as shown in Fig. 6, an HSI cube ($w = 9$) can be seen as a center pixel and four circles in the CP pattern. All pixels are divided into nine regions in spatial dimension, and the center pixel belongs to the first region "R0" (marked in red pixel). The AP operations are

TABLE I
CONFIGURATION OF POOL SIZE AND STRIDE OF THE CP LAYER IN THE CENTRALFORMER MODEL

	Pool size	Strides ($\delta = 1$)
Input width (number of circles)	9 (4)	
R0	(1, 1, 2)	(1, 1, 2)
R1, R3, R5, R7	(2, 2, 2)	
R2, R6	(2, 1, 2)	
R4, R8	(1, 2, 2)	
Output width (number of circles)	7 (3)	

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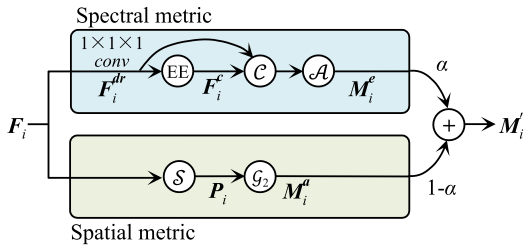


Fig. 7. Architecture of the ARE module, where "EE," "$\mathcal{C}$," "$\mathcal{A}$," "$\mathcal{S}$," and "$\mathcal{G}_2$" indicate extraction and expansion, cosine distance, alignment, discrete sampling, and 2-D Gaussian function, respectively.

conducted in the surrounding regions (R1–R8) of the center pixel. The number of circles can be reduced to proper values with different spatial pool sizes and spatial strides δ. However, the center pixel (R0) is not involved in this pooling process. This is to remain the original feature responses and consolidate the core status of the center pixel in the relevant areas. Thus, compared with other pooling strategies, the CP layer can preserve the characteristics of relevant areas effectively.

The CP layer receives the output $\boldsymbol{F}_i''$ of FFP and produces the dimension-reduced features $\boldsymbol{F}_{i+1} \in \mathbb{R}^{h_i \times w_{i+1} \times w_{i+1} \times b_{i+1}}$

$$\boldsymbol{F}_{i+1} = \mathrm{CP}\left(\boldsymbol{F}_i''\right). \tag{8}$$

The specific setting of pool size (row, column, and band) and stride of the CP layer is given in Table I. In practice, the pooling operation is conducted only in spectral dimension when there is one circle in feature maps, i.e., width $w = 3$.

C. ARE Module

An HSI cube is composed of the center pixel and many neighboring pixels. The effects of a neighboring pixel on improving the attention capture and discrimination of features generally depend on the relevance between this pixel and the center pixel. Previous studies [44], [56] evaluated the relevance map via the spectral metric merely, which ignores the spatial associations and may deviate the estimation results. In this work, the ARE module considers both spectral and spatial affinities between the center pixel and its neighborhoods and fuses them adaptively to get more accurate evaluations. As shown in Fig. 7, the ARE module generates the relevance map $\boldsymbol{M}_i' \in \mathbb{R}^{1 \times w_i \times w_i \times 1}$ from the input $\boldsymbol{F}_i \in \mathbb{R}^{h_{i-1} \times w_i \times w_i \times b_i}$ of the ith central encoder.

In the spectral part, the center pixel is first extracted from the channel-reduced feature map $\boldsymbol{F}_i^{dr} \in \mathbb{R}^{1 \times w_i \times w_i \times b_i}$ and expanded to the center cube $\boldsymbol{F}_i^{c} \in \mathbb{R}^{1 \times w_i \times w_i \times b_i}$ with the "EE" operator. The classic cosine distance is then utilized to measure the normalized spectral angles $\boldsymbol{M}_i^{e} \in \mathbb{R}^{1 \times w_i \times w_i \times 1}$ between the center pixel and its neighboring pixels as follows:

$$\boldsymbol{M}_i^{e} = \mathcal{A}\left(\frac{\boldsymbol{F}_i^{c} \odot \boldsymbol{F}_i^{dr}}{\sqrt{\boldsymbol{F}_i^{c} \odot \boldsymbol{F}_i^{c}} \cdot \sqrt{\boldsymbol{F}_i^{dr} \odot \boldsymbol{F}_i^{dr}}}\right) \tag{9}$$

where "$\odot$" indicates the dot-product between two pixels. The alignment operation "$\mathcal{A}$" is used to linearly transform the range of values from $[-1, 1]$ to $[0, 1]$.

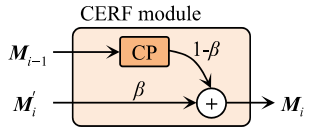


Fig. 8. Architecture of the CERF module.

In the spatial part, the spatial correlations between the center pixel and other pixels $\boldsymbol{M}_i^{a} \in \mathbb{R}^{1 \times w_i \times w_i \times 1}$ are evaluated with 2-D Gaussian function $\mathcal{G}_2$. This process can be described as

$$\boldsymbol{M}_i^{a} = e^{-\mathcal{S}(\boldsymbol{F}_i)} = e^{-\frac{\boldsymbol{P}_i}{2}} = e^{-\frac{\boldsymbol{P}_i^{x} + \boldsymbol{P}_i^{y}}{2}} \tag{10}$$

where $\mathcal{S}$ denotes the discrete sampling. It is the square of the distances $\boldsymbol{P}_i \in \mathbb{R}^{w_i \times w_i}$ between the center pixel and its neighborhoods at the axes of column and row, i.e., $\boldsymbol{P}_i^{x} \in \mathbb{R}^{1 \times w_i}$ and $\boldsymbol{P}_i^{y} \in \mathbb{R}^{1 \times w_i}$.

After preparing two types of correlations, a learnable coefficient α is used to fuse them to the relevance map adaptively

$$\boldsymbol{M}_i' = \alpha \cdot \boldsymbol{M}_i^{e} + (1 - \alpha) \cdot \boldsymbol{M}_i^{a} \tag{11}$$

where the range of α is $[0, 1]$. By replacing the key set with this relevance map, the relevant pixels will be emphasized, while the interfering pixels will be suppressed during the attention deduction and feature extraction of the ith central encoder.

D. CERF Module

The CERF module aims to strengthen the associations of the relevance maps in different levels of features, which can enhance the reliability of relevance maps. Generally speaking, the relevance map extracted by the ARE module represents the pixels related to the center pixel in the current level of features. However, the estimation of these pixels will have deviated seriously when the network becomes deeper [44].

Therefore, as shown in Fig. 8, the current relevance map $\boldsymbol{M}_i' \in \mathbb{R}^{1 \times w_i \times w_i \times 1}$ at deep location and the previous one $\boldsymbol{M}_{i-1} \in \mathbb{R}^{1 \times w_{i-1} \times w_{i-1} \times 1}$ at relatively shallow location are fused adaptively to take previous relevance into account. The fused relevance map $\boldsymbol{M}_i \in \mathbb{R}^{1 \times w_i \times w_i \times 1}$ can be obtained by

$$\boldsymbol{M}_i = \beta \cdot \boldsymbol{M}_i' + (1 - \beta) \cdot \mathrm{CP}(\boldsymbol{M}_{i-1}) \tag{12}$$

where "β" is a learnable coefficient to adjust the weights of different depths of relevance maps. Meanwhile, considering the different spatial sizes of both relevance maps, the CP layer is also exploited to well keep the completeness of previous relevance $\boldsymbol{M}_{i-1}$ and reduce the spatial size. With (12), the deep relevance maps can be effectively improved by the shallow relevance maps, even the first one $\boldsymbol{M}_1$.

III. Experiments

In this section, datasets and experimental configurations are interpreted first. Then, parameter setting, ablation study, and the comparison with other methods are presented. Finally, the effects of sample width and training proportion on classification accuracy are discussed.

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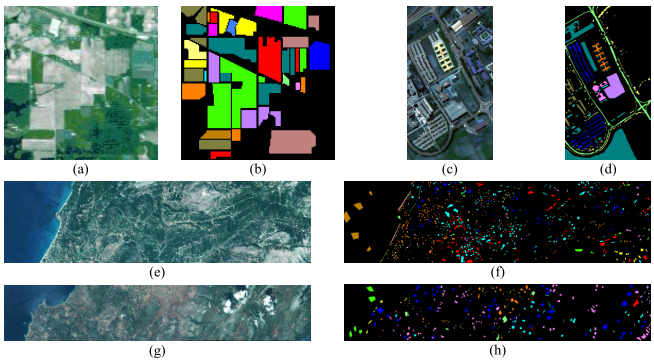


Fig. 9. False-color images and GT maps of four datasets. (a) and (b) Indian Pines. (c) and (d) Pavia University. (e) and (f) Loukia. and (g) and (h) Dioni.

TABLE II

DETAILS OF FOUR DATASETS, WHERE "N_b" AND "N_c" DENOTE THE NUMBER OF BANDS AND THE NUMBER OF CLASSES, RESPECTIVELY

Data Set	Sensor	Year	Size	N_b	N_c
Indian Pines	AVIRIS	1992	145×145	200	16
Pavia University	ROSIS	2002	610×340	102	9
Loukia	Hyperion	2018	945×249	176	14
Dioni	Hyperion	2018	1376×250	176	12

TABLE III

NUMBERS OF SAMPLES OF THE INDIAN PINES DATASET

No.	Land-cover type	Training	Validation	Test
1	Alfalfa	2	2	42
2	Corn-notill	71	71	1286
3	Corn-mintill	41	42	747
4	Corn	12	12	213
5	Grass-pasture	24	24	435
6	Grass-trees	37	37	656
7	Grass-pasture-mowed	1	1	26
8	Hay-windrowed	24	24	430
9	Oats	1	1	18
10	Soybean-notill	49	49	874
11	Soybean-mintill	123	123	2209
12	Soybean-clean	30	30	533
13	Wheat	10	10	185
14	Woods	63	63	1139
15	Building-Grass-Trees-Drives	19	19	348
16	Stone-Steel-Towers	5	5	83
	Total	512	513	9224

TABLE IV

NUMBERS OF SAMPLES OF THE PAVIA UNIVERSITY DATASET

No.	Land-cover type	Training	Validation	Test
1	Asphalt	132	332	6167
2	Meadows	373	933	17343
3	Gravel	42	105	1952
4	Trees	61	153	2850
5	Painted metal sheets	27	67	1251
6	Bare Soil	100	251	4678
7	Bitumen	27	67	1236
8	Self-Blocking Bricks	74	184	3424
9	Shadows	19	47	881
	Total	855	2139	39782

TABLE V

NUMBERS OF SAMPLES OF THE LOUKIA DATASET

No.	Land-cover type	Training	Validation	Test
1	Dense urban fabric	14	15	259
2	Mineral extraction sites	3	3	61
3	Non irrigated arable land	27	27	488
4	Fruit trees	4	4	71
5	Olive groves	70	70	1261
6	Broad-leaved forest	11	11	201
7	Coniferous forest	25	25	450
8	Mixed forest	54	54	964
9	Dense sclerophyllous vegetation	190	190	3413
10	Sparse sclerophyllous vegetation	140	140	2523
11	Sparcely vegetated areas	20	20	364
12	Rocks and sand	24	24	439
13	Water	70	70	1253
14	Coastal water	23	23	405
	Total	675	676	12152

TABLE VI

NUMBERS OF SAMPLES OF THE DIONI DATASET

No.	Land-cover type	Training	Validation	Test
1	Dense urban fabric	63	63	1136
2	Mineral extraction sites	10	10	184
3	Non irrigated arable land	31	31	552
4	Fruit trees	7	7	136
5	Olive groves	88	88	1592
6	Broad-leaved forest	18	18	325
7	Coniferous forest	252	252	4531
8	Mixed forest	319	319	5736
9	Dense sclerophyllous vegetation	88	88	1578
10	Sparse sclerophyllous vegetation	25	25	442
11	Sparcely vegetated areas	80	81	1451
12	Rocks and sand	20	20	358
	Total	1001	1002	18021

A. Datasets

To verify the effectiveness of the proposed model, four HSI datasets [61], [62], including Indian Pines, Pavia University, Loukia, and Dioni, are considered. The properties of them are listed in Table II. Fig. 9 shows false-color images of four datasets and corresponding ground-truth (GT) maps. As shown in Tables III–VI, (5%, 5%, 90%), (2%, 5%, 93%), (5%, 5%, 90%), and (5%, 5%, 90%) of all samples are selected for training, validation, and test sets of four datasets, respectively.

To evaluate the classification performances of the proposed model and other methods, four common metrics, including the recall rate of class, overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ), are used.

B. Implementation Details

In the phase of data preprocessing, the values of each dataset were normalized into the range from 0 to 1. The coefficients α and β of ARE and CERF modules were both initialized to 0.5. The *Xavier* normal distribution [63] was employed to initialize all layers of models. During back propagation, the training error between the true labels $\boldsymbol{Y}' \in \mathbb{R}^{bs \times N}$ and the predicted labels $\boldsymbol{Y} \in \mathbb{R}^{bs \times N}$ in a minibatch was measured by the classic cross-entropy loss function.

The batch size and the number of epochs were set to 32 and 200, respectively. The RMSprop [64] optimizer was adopted to optimize the whole network, where (*learning rate, beta1, beta2*) are set to (0.001, 0.9, 0.999). The learning rate decreased by half every 40 epochs. All experiments were performed on a computer with an Intel Xeon[1] W-11855M CPU at 3.20 GHz × 12 with 64-GB RAM and an NVIDIA RTX A5000 Laptop GPU with 16-GB RAM. The operating system is the Windows 11 Pro for Workstations. The DL frameworks contain Tensorflow 2.6.0, Keras 2.6.0, and Python 3.9.16.

C. Parameter Setting

1) Expansion Ratio of the SSLR Module: The influence of the expansion ratio of the SSLR module, i.e., the ratio of the

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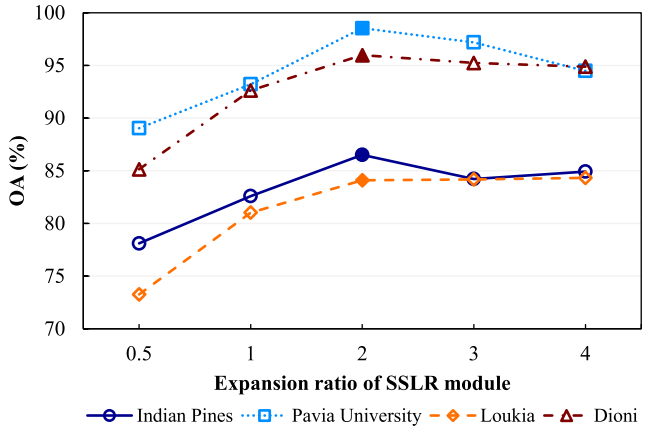


Fig. 10. OAs (%) of the proposed model with different expansion ratios of the SSLR modules, where solid markers denote the optimal options.

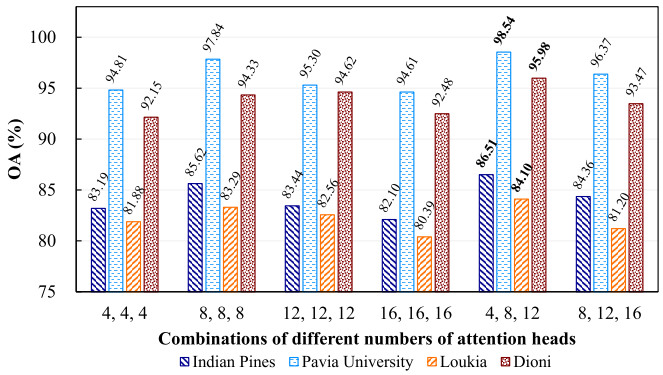


Fig. 11. OAs (%) of the proposed model with different numbers of attention heads.

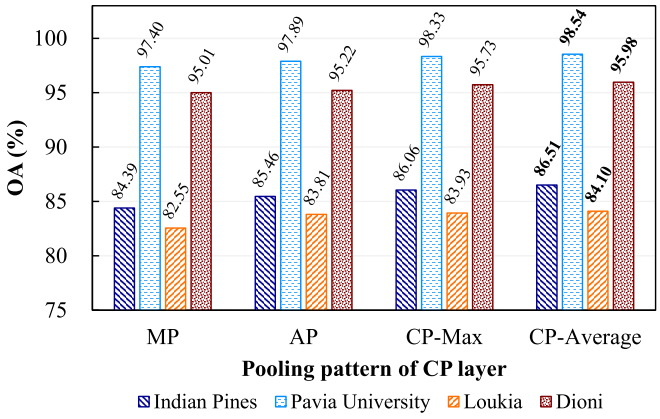


Fig. 12. OAs (%) of the proposed model with different pooling layers.

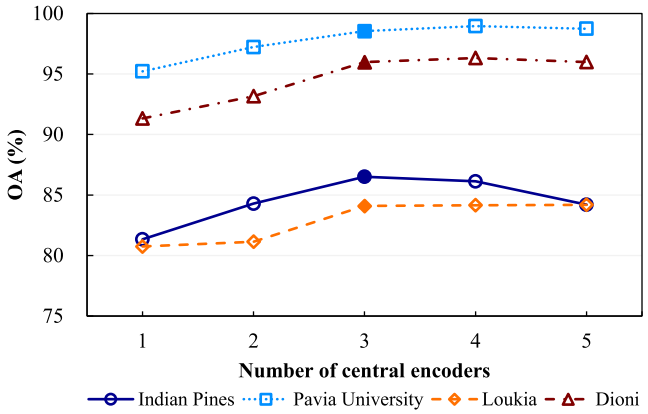


Fig. 13. OAs (%) of the proposed model with different numbers of central encoders, where solid markers denote the optimal options.

number of kernels k_{exp} of PWConv3D or BWConv3D layers to the number of channels of input $\boldsymbol{F}_i^{ci}$, on the expressiveness of local spectral–spatial features cannot be ignored. As shown in Fig. 10, the OAs of the proposed CentralFormer model on four datasets rise first and reduce when higher expansion ratios are used. The setting of a half expansion ratio results in the worst accuracy because the loss of crucial local features may occur during compression. Compared with the setting of no expansion, the OAs of the models with the latter three expansion ratios are all higher, which reveals the rationality of the SSLR module. However, the continued increase in the expansion ratio leads to a decrease in tendency, which is usually related to feature redundancy. Thus, the optimal expansion ratio is set to 2 for all datasets.

2) Number of Attention Heads: The CMHA module is in charge of acquiring the global discriminating features. The number of attention heads has a great effect on attention patterns and the quality of those global features. To reach higher classification accuracy, the numbers of attention heads in three central encoders are set to six combinations {(4, 4, 4), (8, 8, 8), (12, 12, 12), (16, 16, 16), (4, 8, 12), (8, 12, 16)}, respectively. As shown in Fig. 11, among the former four combinations that belong to the fixed-width architecture, the second one (8, 8, 8) and the third one (12, 12, 12) reach the highest OAs on the former three datasets and Dioni datasets, respectively. This inconsistency is caused by that more training samples of the Dioni dataset promote the optimization of parameters. However, the OAs of the fifth combination (4, 8, 12) are the best on four datasets compared with other combinations, which proves the superiority of growing-width architecture [27]. The last combination (8, 12, 16) cannot achieve higher classification accuracy probably due to the limited samples. Therefore, the optimal number of attention heads is set to (4, 8, 12) for the proposed model.

3) Different Types of Pooling Layers: The CP layer of the central encoder is to reduce the circles and bands of feature maps while preserving the characteristics of relevant areas. To demonstrate the necessity, four kinds of pooling layers, including MP, AP, and CP in which MP is conducted in each region (CP-Max) and CP in which AP is conducted in each region (CP-Average), are installed in the proposed model. It can be seen from Fig. 12 that the OAs of CP-Max and CP-Average patterns are higher than those of regular MP and AP patterns, which confirms the effectiveness of CP. Compared with the CP-Max pattern, the CP-Average pattern protects more useful semantic information instead of abandoning them. Hence, it is adopted in the CentralFormer model to acquire the highest classification accuracy.

4) Number of Central Encoders: The central encoder aims to extract the discriminating local and global features for HSI classification. Its number can influence the depth and representation ability of features. In this part, different numbers

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TABLE VII

ABLATION STUDY OF THE PROPOSED MODEL ON THE INDIAN PINES (IP), PAVIA UNIVERSITY (PU), LOUKIA (LK), AND DIONI (DN) DATASETS

No.	Schemes	IP	PU	LK	DN
1	SSLR	79.86	94.48	78.49	91.57
2	MHA	81.66	95.69	80.23	92.09
3	CMHA	83.58	96.51	81.50	93.81
4	CMHA + SSLR	83.05	95.87	80.66	93.11
5	SSLR + CMHA	84.13	97.46	82.28	94.77
6	SSLR + CMHA + FFP	85.08	97.89	82.96	94.83
7	SSLR + CP	80.06	94.37	78.74	91.89
8	CMHA + CP	83.87	96.22	81.77	94.05
9	Central encoder (SSLR + CMHA + FFP + CP)	85.22	97.85	83.19	95.05
10	Central encoder + ARE (Spectral metric only)	85.84	98.06	83.66	95.43
11	Central encoder + ARE (Spatial metric only)	86.01	97.86	83.59	95.23
12	Central encoder + ARE	86.11	98.22	83.81	95.62
13	Central encoder + ARE + CERF	**86.51**	**98.54**	**84.10**	**95.98**

{1, 2, 3, 4, 5} of central encoders are set to explore proper architectures for four datasets. As shown in Fig. 13, the OAs of models on four datasets increase when the number of central encoders is no more than 3. It is the multilevel architecture that enhances deep feature representation. However, obvious decreases in the OAs of models on the Indian Pines dataset can be seen when there are more than three central encoders deployed. In contrast, the OAs of models on other datasets remain stable. This is because the number of parameters in additional encoders and insufficient training samples limit jointly the extraction of deeper features. Therefore, the numbers of central encoders of models for all datasets are set to 3.

D. Ablation Study

The proposed CentralFormer model is mainly composed of the central encoders, ARE modules, and CERF modules. In this section, the rationality of three components is studied with 13 combinations shown in Table VII. Schemes 1–9 are all possible combinations of the components of the central encoder, including the SSLR module, the CMHA module, the FFP, and the CP layer. It can be observed that the OAs of the model with global features (Scheme 3) are better than those with local features (Scheme 1), which is caused by the inherent different between convolution and global attention. SSLR and CMHA modules are deployed in sequence to take into account both local and global features. However, Scheme 5 reaches higher OAs than Scheme 4 because local features are suitable for attention deduction, while global features are beneficial to classification. The effectiveness of the centralization ideology of the CMHA module is also verified from the comparison between Schemes 2 and 3. After the FFP is considered (Scheme 6), the fused features possess stronger representation ability and elevate the classification accuracy. Compared with corresponding nonpooling combinations (Schemes 1, 3, and 6), Schemes 7–9 improve further corresponding OAs on IP, LK, and DN datasets. The reason why there are slight decreases of Schemes 7–9 on the PU dataset after introducing the CP layer is mainly related to the smaller sample width, which limits the number of conductions of CP. However, it still can reveal that the CP layer is helpful in maintaining the relevant features during feature reduction. Schemes 10–12 explore the effects of spectral and spatial metrics on classification accuracy. It can be seen that the ARE module that adopts both spectral and spatial metrics is the best choice compared with Schemes 10 and 11. The last scheme studies the importance of the CERF module, which is to fuse the relevance maps between different levels of features. Compared with Scheme 12, the OAs of the model on four datasets receive different degrees of increase. This confirms the necessity of ensuring the reliability of the relevance map and the effectiveness of the CERF module.

E. Classification Results

To confirm the effectiveness of the proposed CentralFormer model, ten methods, including SVM [6], k-NN [7], the spectral–spatial residual network (SSRN) [32], the spectral–spatial dense network (SSDN) [34], the fused attention network (FAN) [43], the spectral former (SF) [48], the central attention network (CAN) [49], the masked spectral–spatial transformer (MSST) [52], the center-masked transformer (CMT) [54], and the center attention transformer (CAT) [55], were considered for comparison. The original architectures of these methods were adopted according to corresponding articles. To ensure the fairness of results, all methods shared the same training set, validation set, and test set. All experiments were repeated ten times, and the average results and standard deviations of different methods were recorded. Sections III-E1–III-E3 present the quantitative results, qualitative results, and complexity analysis of these methods.

1) Quantitative Evaluation: Tables VIII–XI report the recall rates of classes, OAs, AAs, and κ's of different methods on four datasets. The bold values are the best result of corresponding rows. Four conclusions can be derived from these tables.

First, SVM and KNN, as traditional machine learning algorithms, achieve unsatisfying classification performances on all datasets. The recall rates of some classes, such as the fourth class on the Dioni dataset, are even zero. Meanwhile, κ's of SVM and KNN are all less than 65%. This means that there are large rooms for the two algorithms to improve the complex spectral–spatial features.

Second, CNN-based models, i.e., SSRN and SSDN, elevate classification results obviously compared with SVM and KNN. SSDN generates more levels of features with dense connections and reaches better classification accuracy than SSRN.

Third, benefit from attention mechanism, FAN, SF, CAN, MSST, CMT, and CAT reach considerably high classification performances. FAN integrates SA into both spectral and spatial residual subnetworks to highlight the important bands and positions. It receives higher classification accuracy than SF. In particular, the OA of FAN is almost 20% higher than that of SF on the Pavia University dataset. The main reason is that the ability of SF to model the long-term spectral dependence is restricted by the fewer bands of the Pavia University dataset. Moreover, the low recalls of some classes of SF on four datasets also cut down its AAs and κ's. Other four spectral–spatial transformer architectures, CAN, MSST, CMT, and CAT, obtain better classification results than FAN. This is because they attach importance to the center pixel to extract more patterns of key information. Specifically, CAN adopts the center pixel as the query pixel for attention

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TABLE VIII

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE INDIAN PINES DATASET WITH 5% OF AVAILABLE LABELED PIXELS

No.	SVM	KNN	SSRN	SSDN	FAN	SF	CAN	MSST	CMT	CAT	Ours
1	18.43±5.09	14.88±7.15	17.86±9.95	17.62±10.33	50.48±11.61	32.14±10.82	43.81±10.82	42.86±15.06	49.29±13.80	46.67±15.76	**54.76±21.08**
2	55.19±4.63	52.30±3.36	60.49±4.67	57.99±6.27	71.70±2.49	66.63±4.07	74.03±1.58	73.98±1.94	73.70±2.43	73.67±2.40	**74.36±4.03**
3	42.40±7.79	39.81±5.54	59.33±6.63	64.11±2.84	66.03±4.28	65.18±4.52	72.98±5.46	73.08±3.53	74.66±7.17	72.61±6.83	**81.28±4.27**
4	18.15±4.01	16.90±4.91	30.19±6.71	37.65±6.15	47.98±5.95	40.33±6.99	50.42±5.23	52.07±5.23	57.00±7.79	52.11±4.65	**71.64±16.24**
5	63.48±7.87	61.19±8.76	76.12±6.24	80.02±4.19	81.68±5.67	76.85±6.74	87.41±2.97	87.75±2.36	87.49±3.57	87.73±2.42	**87.96±2.21**
6	78.42±9.04	75.51±8.11	94.94±1.66	96.91±1.17	92.26±2.34	93.86±3.99	94.83±1.73	95.52±1.32	94.86±2.65	94.91±1.59	**97.26±0.65**
7	28.08±7.31	26.93±8.94	11.15±4.37	12.31±3.77	27.69±7.46	16.15±2.88	**30.38±8.15**	26.54±7.39	25.77±5.45	23.85±3.77	28.08±7.10
8	93.17±3.64	93.78±3.74	97.91±2.08	98.93±0.74	99.47±0.80	98.95±1.02	99.67±0.83	99.61±1.11	99.40±1.18	99.54±1.10	**99.88±0.16**
9	1.67±5.00	1.67±5.00	26.67±13.10	26.11±8.62	37.78±6.48	27.22±7.22	34.44±9.56	28.89±5.98	40.00±5.98	33.33±8.24	**43.33±9.56**
10	52.79±5.11	53.35±5.99	59.14±10.15	63.84±5.30	79.03±2.33	71.55±5.29	**79.27±2.59**	78.26±2.95	78.71±4.33	78.87±4.04	79.10±3.19
11	75.91±2.53	75.00±2.42	73.23±1.91	76.00±2.91	85.60±1.40	80.11±4.67	87.49±1.46	86.42±1.77	86.97±2.12	86.31±1.68	**89.91±1.53**
12	38.07±4.80	35.12±4.03	48.01±6.96	48.18±5.38	63.85±3.97	57.64±4.47	64.56±4.22	64.54±4.84	67.99±4.90	65.57±4.91	**68.67±2.86**
13	97.77±2.91	98.09±2.67	93.35±2.84	95.51±2.81	**98.97±1.17**	97.89±2.20	98.60±1.09	98.65±0.69	98.27±1.23	98.27±1.13	97.25±1.54
14	85.97±5.51	83.35±6.46	**95.71±1.43**	95.09±1.71	89.89±1.80	92.14±3.32	92.07±1.04	92.42±2.01	92.93±2.86	92.71±2.24	95.00±2.11
15	39.16±8.54	35.78±7.71	53.42±5.79	65.63±8.37	73.22±4.67	68.79±5.69	77.73±3.65	80.06±5.96	85.55±6.54	82.85±6.52	**92.36±4.97**
16	21.44±19.58	15.95±8.50	**59.88±14.17**	47.95±14.73	26.27±3.13	32.65±16.96	25.78±4.38	23.74±4.35	26.99±3.54	24.34±3.92	43.38±25.34
OA	64.26±2.60	62.48±2.00	71.17±2.81	73.19±0.76	79.65±1.33	76.18±1.85	81.97±1.18	81.80±0.74	82.65±1.73	81.92±0.85	**85.29±0.72**
AA	50.63±3.36	48.73±2.36	59.84±3.47	61.49±1.24	68.24±1.99	63.63±2.59	69.59±1.63	69.02±1.27	71.22±2.57	69.58±1.57	**75.26±2.59**
$\kappa \times 100$	58.78±3.03	56.74±2.30	66.95±3.32	69.27±0.95	76.69±1.52	72.72±2.13	79.35±1.35	79.18±0.87	80.15±1.99	79.32±1.00	**83.17±0.84**

TABLE IX

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE PAVIA UNIVERSITY DATASET WITH 2% OF AVAILABLE LABELED PIXELS

No.	SVM	KNN	SSRN	SSDN	FAN	SF	CAN	MSST	CMT	CAT	Ours
1	94.05±12.59	69.01±3.11	94.42±2.00	94.87±2.24	97.15±0.65	93.20±2.56	97.16±0.37	96.88±0.41	96.80±0.87	97.27±0.60	**98.32±0.69**
2	98.47±3.22	91.67±0.38	98.09±0.39	98.49±0.19	98.76±0.09	98.80±1.96	99.16±0.35	99.16±0.37	**99.80±0.09**	99.68±0.16	99.71±0.16
3	2.75±6.15	22.78±6.27	80.38±2.47	80.09±2.47	83.40±2.24	10.27±19.72	85.85±3.65	83.80±3.05	85.45±4.64	86.44±2.21	**90.34±2.17**
4	45.90±45.79	84.25±1.55	95.90±1.12	93.44±3.05	91.21±2.68	76.10±6.90	91.48±2.25	91.29±2.64	**96.53±0.72**	95.18±1.71	94.53±1.63
5	93.58±11.61	71.31±3.60	**100.00±0.00**	99.35±0.72	99.74±0.36	97.67±2.49	99.61±0.28	99.60±0.29	99.79±0.16	99.54±0.39	99.80±0.33
6	5.88±9.13	29.53±3.75	90.41±3.58	91.01±3.85	95.40±2.07	13.20±10.65	96.45±1.79	96.16±2.05	**99.04±0.34**	98.11±1.04	98.88±0.96
7	0.01±0.03	4.01±3.93	84.63±2.61	84.24±2.03	86.53±4.03	0.00±0.00	89.81±2.28	88.47±2.43	87.54±4.24	87.72±2.78	**92.95±1.80**
8	22.99±34.28	84.01±4.57	92.74±3.23	92.71±3.05	93.94±1.75	90.85±8.58	95.55±1.22	95.60±1.39	94.97±2.33	95.86±0.54	**96.36±1.54**
9	47.11±47.17	89.73±3.35	98.87±1.22	99.75±0.38	98.67±1.22	52.30±43.37	99.85±0.26	99.85±0.20	**100.00±0.00**	99.89±0.12	99.90±0.27
OA	67.59±4.22	72.87±1.06	94.79±0.42	94.90±0.51	96.06±0.43	77.07±1.23	96.76±0.55	96.53±0.55	97.52±0.30	97.45±0.26	**98.07±0.32**
AA	45.64±10.88	60.70±1.29	92.83±0.67	92.66±0.42	93.87±0.76	59.16±6.15	94.99±0.89	94.54±0.76	95.55±0.89	95.52±0.35	**96.75±0.43**
$\kappa \times 100$	53.12±7.41	63.21±1.55	93.08±0.57	93.22±0.67	94.76±0.58	67.89±2.09	95.69±0.74	95.38±0.73	96.70±0.39	96.61±0.34	**97.44±0.42**

TABLE X

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE LOUKIA DATASET WITH 5% OF AVAILABLE LABELED PIXELS

No.	SVM	KNN	SSRN	SSDN	FAN	SF	CAN	MSST	CMT	CAT	Ours
1	0.83±1.44	3.53±4.83	18.03±1.87	19.49±4.48	24.94±5.89	23.66±5.99	32.24±5.21	29.27±6.36	33.90±4.78	31.32±7.00	**38.44±6.49**
2	3.57±6.19	6.98±13.97	37.70±26.14	34.28±11.47	30.48±13.37	62.63±18.99	75.96±10.52	81.15±5.35	84.92±5.22	80.53±9.98	**85.53±10.01**
3	18.91±18.55	32.32±15.61	56.54±12.42	63.77±5.09	67.86±5.45	68.67±4.16	74.39±1.22	72.85±4.71	79.75±2.36	76.46±4.74	**82.14±2.66**
4	0.68±1.17	0.81±1.62	5.82±3.54	8.63±2.61	18.08±7.91	7.41±2.86	16.90±6.30	12.67±5.97	18.03±7.84	14.79±5.08	**22.59±9.29**
5	8.35±14.47	20.54±26.34	58.67±7.93	69.30±7.75	77.17±8.17	69.88±5.43	81.27±3.71	85.19±2.32	86.54±2.70	83.44±2.97	**88.82±2.18**
6	0.36±0.62	1.45±1.92	20.54±8.29	30.50±7.46	31.34±8.77	33.75±7.18	43.37±3.09	40.25±5.02	46.87±2.66	45.27±4.74	**47.55±8.31**
7	32.48±19.96	24.13±19.90	52.34±1.86	52.26±5.33	53.98±7.06	53.22±3.00	54.45±2.94	58.16±4.38	**61.33±3.60**	56.89±1.93	58.82±2.78
8	23.44±23.49	57.69±10.76	50.39±5.25	58.70±5.30	65.98±4.40	62.42±7.64	67.07±3.70	69.98±4.40	71.93±3.46	70.37±4.13	**72.31±3.52**
9	85.38±10.14	75.59±7.65	79.17±1.55	78.95±1.72	80.62±3.22	81.95±1.86	84.27±0.86	84.52±1.94	85.04±1.95	84.50±1.07	**85.79±1.95**
10	53.36±31.45	73.82±5.59	76.30±1.82	79.29±3.96	78.77±2.68	78.54±3.68	80.77±1.67	83.64±2.61	83.61±0.98	83.14±1.64	**84.52±1.88**
11	10.74±14.63	16.83±22.00	46.96±5.79	46.12±7.53	54.36±5.08	44.55±7.61	52.06±5.50	50.47±4.92	53.96±3.32	55.50±3.23	**62.47±4.28**
12	58.58±35.77	77.44±8.71	86.44±6.79	85.97±4.92	90.02±3.09	88.90±4.07	93.02±1.26	92.67±1.15	**93.76±0.65**	93.42±0.99	93.54±0.59
13	**100.00±0.00**	99.98±0.03	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
14	74.14±42.81	97.29±3.78	99.94±0.10	99.63±0.54	**100.00±0.00**	99.47±0.50	99.96±0.09	99.80±0.24	99.46±0.74	99.82±0.40	99.88±0.20
OA	55.02±7.71	62.45±1.69	71.49±0.49	74.24±1.83	76.80±2.06	75.85±1.54	79.64±1.22	80.84±0.51	82.06±0.49	81.00±1.12	**83.13±0.44**
AA	34.53±8.28	42.03±4.29	56.34±0.93	59.06±1.46	62.40±3.25	62.50±2.60	68.27±2.17	68.62±0.77	71.36±0.85	69.68±1.80	**73.03±1.21**
$\kappa \times 100$	42.77±12.16	53.95±2.35	65.55±0.73	69.03±2.23	72.19±2.51	70.95±1.89	75.58±1.50	77.03±0.60	78.54±0.57	77.23±1.37	**79.81±0.52**

deduction, but other relevant pixels are ignored during the query process. MSST masks some pixels and bands randomly to improve the image reconstruction and enhance the robustness of the model. However, the important characteristics of the relevant pixels will be suppressed. CMT prepares the local-to-global embeddings and controls the accuracy of the reconstruction of the center pixel, which enables the model to learn the correlations between the center pixel and others. CAT generates the spectral–spatial embeddings with the center pixel, the homogeneous region, and the neighborhood region to help the model focus on the center pixel during feature extraction. However, among the four models, CMT obtains the highest classification performances on four datasets because the parameter optimization of CMT is improved by the sample reconstruction, especially for the center pixel.

Finally, different from other attention-based methods, the proposed CentralFormer model adopts the central encoder and the ARE module to emphasize the relevant pixels, including the center pixel, during attention deduction and feature representation. To prevent attention from deviating from the relevant pixels when the network is deep, the CERF module takes into account the peculiarity of the previous relevance map to ensure the precision of the current relevance map. It can be seen that the numbers of the highest recall rates of classes of

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TABLE XI

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE DIONI DATASET WITH 5% OF AVAILABLE LABELED PIXELS

No.	SVM	KNN	SSRN	SSDN	FAN	SF	CAN	MSST	CMT	CAT	Ours
1	9.27±16.05	1.65±2.86	66.19±5.52	67.71±6.65	82.57±3.96	71.67±4.69	84.98±3.47	88.20±0.63	**89.76±2.02**	89.03±1.30	89.75±1.86
2	26.90±27.60	11.28±19.53	28.91±9.73	38.93±17.54	63.00±7.83	43.56±19.55	86.97±4.74	87.39±6.36	**91.22±4.56**	88.48±7.64	88.21±4.59
3	6.66±11.53	9.74±10.97	64.65±8.47	67.59±7.22	78.43±2.53	70.00±9.14	82.43±3.27	82.17±4.83	75.06±3.57	77.43±3.33	**88.46±3.16**
4	0.00±0.00	0.00±0.00	4.05±2.61	4.05±3.50	43.05±16.63	12.41±14.22	62.70±9.42	62.35±4.61	45.47±8.38	59.71±12.00	**66.10±4.95**
5	35.24±13.65	14.21±12.49	73.53±7.40	82.76±8.38	92.58±2.01	81.84±9.26	92.16±1.23	94.41±2.46	95.17±0.81	93.88±1.09	**96.82±1.46**
6	31.08±25.80	18.77±26.20	86.44±3.77	91.37±5.90	97.72±1.56	90.34±6.10	**99.31±0.40**	99.14±0.23	97.49±1.26	98.03±1.11	98.95±0.63
7	84.12±4.62	85.29±2.55	88.37±1.77	90.08±0.26	92.02±0.97	90.06±1.73	91.96±0.51	91.87±0.66	**94.56±0.83**	93.40±0.97	94.54±0.80
8	88.83±2.82	91.44±1.15	96.03±0.88	96.74±0.65	96.28±1.12	96.35±0.85	96.44±0.53	97.05±0.60	96.99±0.34	97.04±0.48	**97.45±0.21**
9	71.63±3.95	76.30±5.42	74.88±1.21	71.27±4.01	81.19±4.24	72.15±3.65	84.16±3.53	88.61±1.22	**93.05±1.08**	90.00±2.01	92.29±1.92
10	30.43±34.53	25.23±26.77	84.53±3.39	78.03±8.99	93.52±3.34	84.77±3.08	97.34±1.92	97.74±1.76	96.08±1.59	95.70±1.99	**98.96±0.62**
11	99.93±0.12	99.71±0.51	99.51±0.53	**100.00±0.00**	**100.00±0.00**	99.75±0.45	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
12	41.13±41.66	16.13±27.94	50.34±4.15	60.97±5.10	71.50±12.80	58.66±11.47	80.82±20.39	94.97±3.47	95.16±3.24	**96.20±3.30**	96.03±2.01
OA	70.04±2.58	68.31±1.52	84.96±1.23	86.58±1.12	91.17±0.57	86.96±1.97	92.41±1.12	93.65±0.51	94.49±0.23	93.96±0.25	**95.36±0.49**
AA	43.77±8.07	37.48±3.10	68.12±1.64	70.79±1.76	82.66±3.13	72.63±4.87	88.28±3.51	90.33±1.19	89.17±1.49	89.91±1.42	**92.30±0.61**
$\kappa \times 100$	61.85±3.58	59.22±2.16	81.09±1.56	83.13±1.43	88.99±0.73	83.63±2.51	90.54±1.42	92.10±0.63	93.15±0.29	92.49±0.31	**94.23±0.61**

TABLE XII

STATISTICAL SIGNIFICANCES (p) IN TERMS OF OA BETWEEN THE PROPOSED MODEL AND OTHER METHODS ON FOUR DATASETS

	SVM	KNN	SSRN	SSDN	FAN	SF	CAN	MSST	CMT	CAT	Ours
Indian Pines	1.32E-09	4.94E-10	5.24E-07	6.96E-11	6.92E-07	4.92E-07	0.001844	4.18E-05	0.009306	0.000108	—
Pavia University	3.72E-10	1.02E-13	5.79E-09	3.32E-08	1.84E-06	3.30E-12	0.000212	4.51E-06	0.012400	0.005075	—
Loukia	1.00E-06	3.47E-11	5.42E-11	7.88E-07	2.09E-05	8.53E-07	1.28E-05	0.000264	0.025767	0.002626	—
Dioni	8.90E-11	5.28E-12	8.21E-11	1.18E-09	3.14E-08	1.62E-07	5.19E-05	0.000411	0.011194	0.001151	—

the proposed CentralFormer model on four datasets are the most, such as the third and 15th classes of the Indian Pines dataset, the third and seventh classes of the Pavia University dataset, the 11th class of the Loukia dataset, and the third and fourth classes of the Dioni dataset. Meanwhile, OA, AA, and κ of the proposed model are also the highest on each dataset. Therefore, it can be sure that the proposal represents the discriminating spectral–spatial features and receives superior classification performances compared with others.

To demonstrate the effectiveness of the proposed model, the statistical significance in terms of OA was analyzed with the paired t-test method. Table XII reports the probability (p) that the proposed model is associated with other methods within ten experiments. It can be seen that the relatively high correlation probabilities of CMT are still less than the common threshold ($\gamma = 0.05$). Hence, there are salient differences between the proposed model and all compared methods.

2) Qualitative Evaluation: Figs. 14–17 show the best classification maps of different methods on four datasets. It can be observed intuitively that the classification maps of SVM, KNN, SSRN, and SSDN contain more speckle noise compared with others. After considering the attention mechanism, the recognition of both borders and regions of different classes is improved to a great extent, especially for the classes with more samples, such as the second and 11th of the Indian Pines dataset and the second and sixth classes of the Pavia University dataset. When it comes to the local details, first, the classification results of the third and 15th classes of Fig. 14, the third and fourth classes of Fig. 15 (marked with cyan rectangles), and the amplified parts of Figs. 16 and 17 (marked with red rectangles) predicted by the proposed model are more similar to corresponding GT maps. Second, it can be seen from the results in these marked rectangles that the proposed model excels at recognizing the pixels on the borders of land covers compared with other methods. The boundary pixels containing different spectral characteristics of multiple classes are harmful to feature extraction. With the ARE module, the proposed model can remain the relevant areas for attention deduction and identification. Unfortunately, other methods even failed to identify the pixels within the region of land covers. Third, when there are limited samples, the classification results of all methods are unsatisfactory (the seventh and ninth classes of the Indian Pines dataset, marked with cyan ellipses in Fig. 14). The reason is that the feature representation of these classes in high-dimensional space cannot be modeled effectively with a few samples. In a word, from the global perspective, the proposed CentralFormer model obtains more accurate classification maps than other methods.

3) Complexity Analyses: To interpret the efficiency of the proposed model, the number of parameters, floating point operations (FLOPs), and training and test times of different methods on four datasets are reported in Table XIII. It can be seen that the nonneural network algorithms, i.e., SVM and KNN, spent quite a short time on training, but their test procedures were lengthy. This is because the computational complexity of both algorithms is sensitive to the number of samples and feature dimensions. On the contrary, other DL-based models generally adopted convolutional layers, pooling layers, and fully connected layers for feature extraction and dimensionality reduction, which cost far more time to train than test. SSRN and SSDN, as the CNN-based models, own lots of parameters and high FLOPs due to abundant residual connections and dense connections. Even so, their time consumption was still less than most attention-based models, which contains many high-dimensional matrix operations. Among six attention-based models for comparison, FAN with a few SA models only finished the training and test procedures efficiently. SF, CAN, MSST, CMT, and CAT applied a series of multihead attention modules to refine features, which consumed a considerable long time for optimization. Different forms of embedding generation methods

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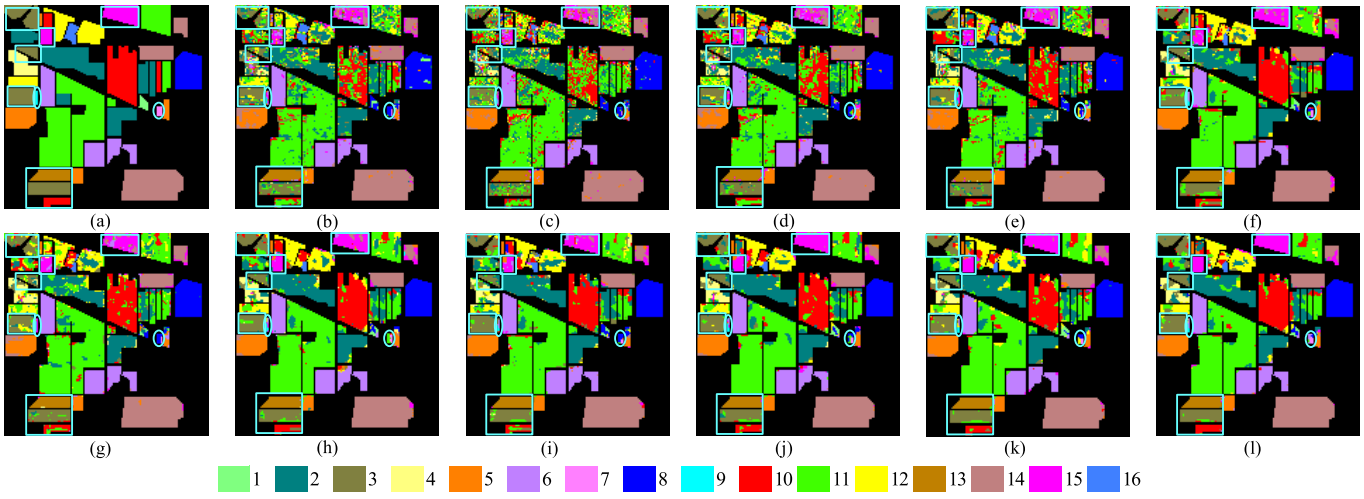


Fig. 14. Classification maps of different methods on the Indian Pines dataset. (a) GT. (b) SVM (68.37%). (c) KNN (64.42%). (d) SSRN (72.25%). (e) SSDN (73.92%). (f) FAN (80.87%). (g) SF (78.76%). (h) CAN (83.97%). (i) MSST (82.89%). (j) CMT (84.04%). (k) CAT (83.26%). (l) CentralFormer (86.51%).

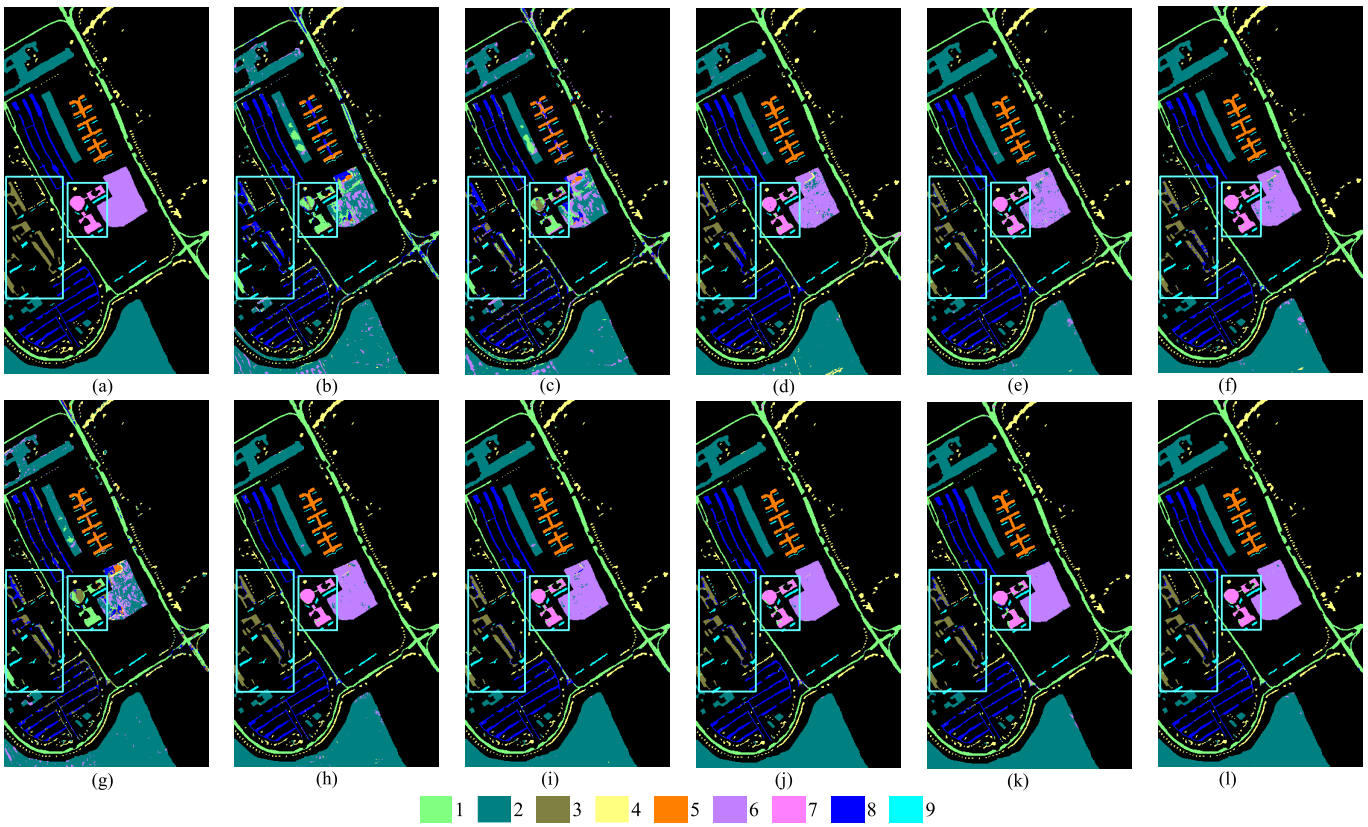


Fig. 15. Classification maps of different methods on the Pavia University dataset. (a) GT. (b) SVM (71.81%). (c) KNN (73.93%). (d) SSRN (95.29%). (e) SSDN (95.39%). (f) FAN (96.47%). (g) SF (78.45%). (h) CAN (97.27%). (i) MSST (97.04%). (j) CMT (97.75%). (k) CAT (97.72%). (l) CentralFormer (98.54%).

also slowed down the convergences of MSST, CMT, and CAT. The proposed CentralFormer model adopted three lightweight central encoders merely to extract the local and global features. More importantly, the CP layers were deployed between encoders to reduce the dimension of relevant features. As a result, four metrics of the proposed model on all datasets were less than other transformer architectures. Compared with other methods, the efficiency of the CentralFormer model is worth approving.

F. Influence of Sample Width

Sample width [65], an important property of the HSI cube, determines the number of pixels for classification. Samples with larger widths generally contain more interfering pixels, which hampers feature extraction. To observe the influence of sample width on the classification accuracy of different methods, different widths {3, 5, 7, 9, 11, 13, 15} of samples were prepared. It can be seen from Fig. 18 that the OAs of

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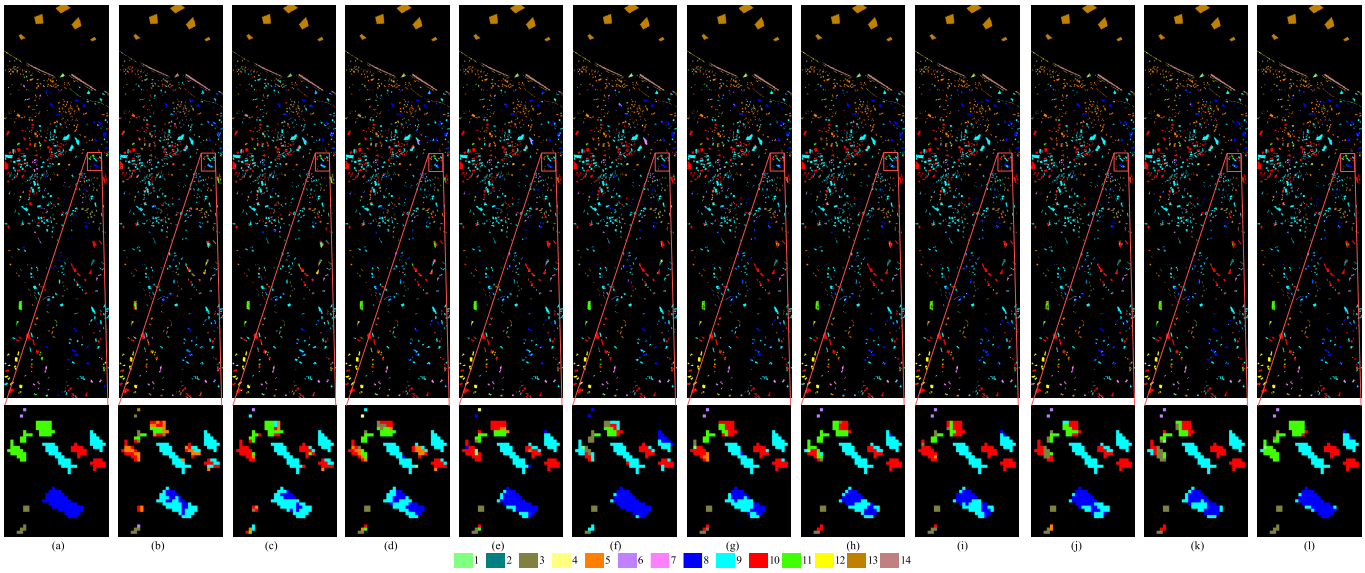


Fig. 16. Classification maps of different methods on the Loukia dataset. (a) GT. (b) SVM (60.23%). (c) KNN (65.01%). (d) SSRN (72.20%). (e) SSDN (76.55%). (f) FAN (78.80%). (g) SF (77.11%). (h) CAN (80.64%). (i) MSST (81.86%). (j) CMT (82.85%). (k) CAT (82.23%). (l) CentralFormer (84.10%).

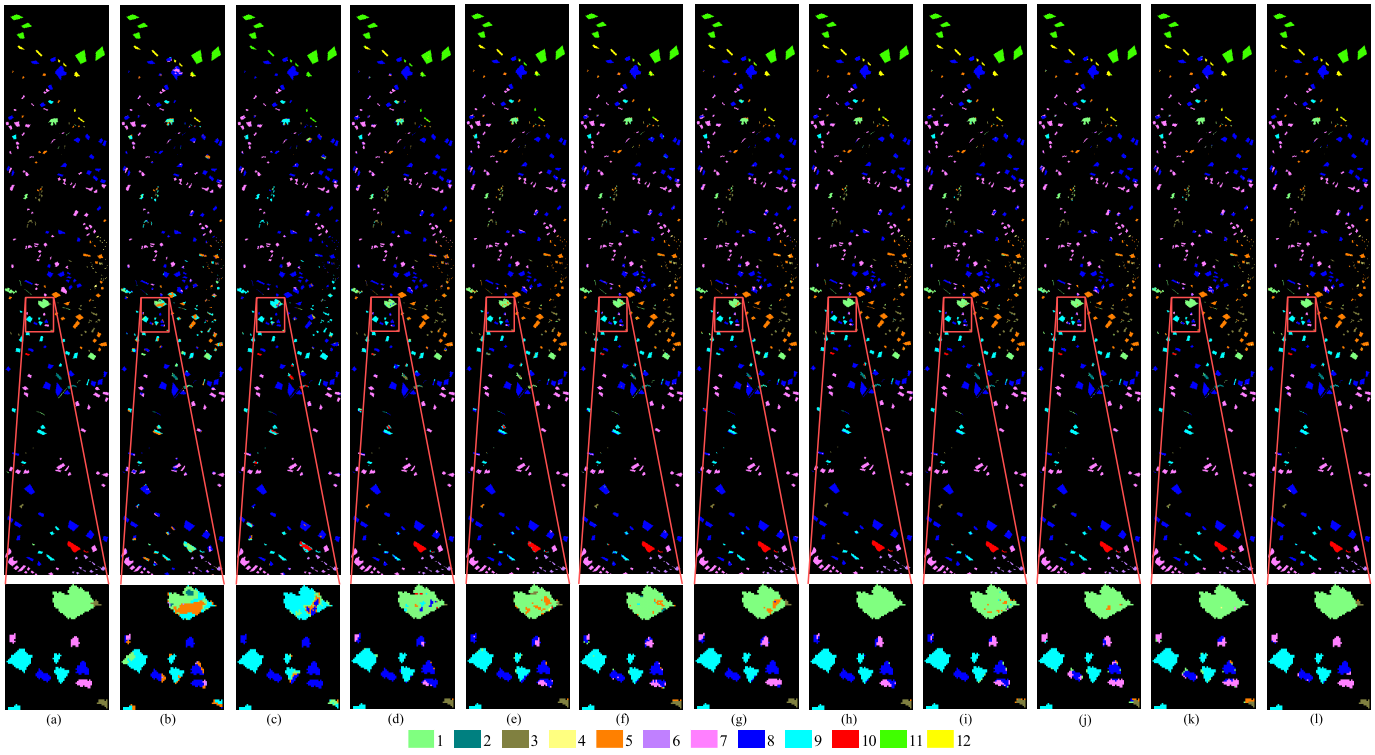


Fig. 17. Classification maps of different methods on the Dioni dataset. (a) GT. (b) SVM (73.03%). (c) KNN (69.75%). (d) SSRN (86.82%). (e) SSDN (88.35%). (f) FAN (91.97%). (g) SF (90.64%). (h) CAN (93.72%). (i) MSST (94.21%). (j) CMT (94.77%). (k) CAT (94.20%). (l) CentralFormer (95.98%).

most methods on four datasets increase first and then fluctuate when sample width becomes larger. SVM, KNN, SSRN, and SSDN achieve relatively low accuracy (orange lines) compared with other attention-based methods (deep blue lines). This is the attention mechanism that helps models highlight the crucial areas in wider samples. However, benefiting from the ARE and CERF modules, the effective spatial areas can be preserved during different stages of feature extraction. Hence, the proposed model (red solid lines) realizes satisfactory classification results on both smaller and larger samples. Considering accuracy and efficiency, the optimal sample widths of four datasets are set to 11, 5, 7, and 5, respectively.

G. Influence of Training Proportion

To clarify the abilities of different models to cope with the issue of limited samples, different percentages of samples from four datasets were selected for training. As shown in Fig. 19, the OAs of different methods all keep the growing tendency with the increase in the training proportion. It can

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TABLE XIII
NUMBER OF PARAMETERS (PARA.), FLOPS, AND TIME CONSUMPTION (SECOND) OF DIFFERENT METHODS ON FOUR DATASETS

		SVM	KNN	SSRN	SSDN	FAN	SF	CAN	MSST	CMT	CAT	Ours
Indian Pines	Para.	—	—	698.88K	866.12K	29.95K	98.94K	58.36K	163.68K	52.20K	96.34K	41.76K
	FLOPs	—	—	3.38M	4.23M	3.81M	1.98M	5.98M	3.62M	1.76M	2.29M	1.20M
	training	3.45	5.41	224.10	382.78	375.00	538.74	496.98	727.86	631.29	675.51	489.35
	test	15.20	20.84	5.33	9.59	8.45	11.23	11.17	18.22	14.05	14.31	9.36
Pavia University	Para.	—	—	55.03K	866.11K	4.51K	92.34K	46.91K	79.14K	13.28K	65.74K	5.97K
	FLOPs	—	—	4.48M	6.95M	2.47M	1.79M	6.03M	2.69M	1.33M	1.88M	1.11M
	training	2.78	3.98	256.99	408.06	397.65	567.16	522.54	744.04	643.77	681.90	510.00
	test	30.60	41.09	11.33	22.35	15.47	22.31	22.23	23.20	25.55	24.48	19.46
Loukia	Para.	—	—	66.17K	866.13K	19.63K	94.20K	61.30K	98.27K	24.36K	51.36K	12.85K
	FLOPs	—	—	5.80M	7.68M	5.11M	1.61M	6.21M	2.85M	1.42M	1.98M	1.23M
	training	2.55	3.62	273.15	358.33	363.15	454.49	390.37	597.46	489.61	495.18	354.44
	test	17.79	15.86	6.16	8.84	7.86	10.32	8.52	13.53	12.64	11.25	8.21
Dioni	Para.	—	—	66.17K	866.13K	15.38K	92.53K	59.93K	93.18K	14.68K	50.68K	11.26K
	FLOPs	—	—	7.13M	10.40M	7.05M	1.43M	7.60M	2.66M	1.39M	1.76M	1.25M
	training	5.82	5.10	278.75	386.89	383.88	496.35	444.14	643.57	477.63	533.47	358.64
	test	23.69	21.52	5.46	6.46	7.31	9.17	9.39	13.02	11.19	11.02	8.12

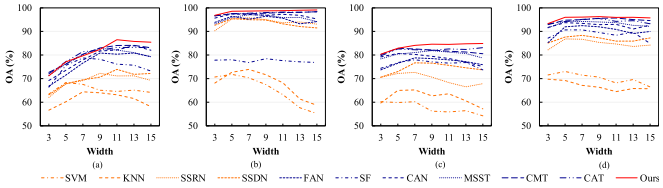


Fig. 18. OAs (%) of different models with different widths of samples. (a) Indian Pines. (b) Pavia University. (c) Loukia. (d) Dioni.

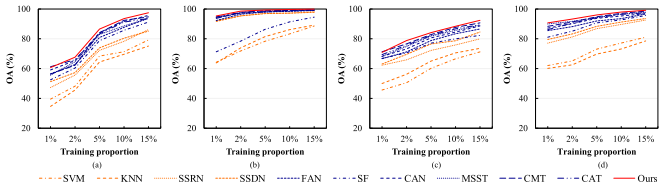


Fig. 19. OAs (%) of different models with different proportions of training samples. (a) Indian Pines. (b) Pavia University. (c) Loukia. (d) Dioni.

be observed that the gaps between the lowest OAs and the highest OAs of most methods on the Indian Pines and Loukia datasets are more serious than those of the other two datasets because of the difference between the numbers of samples of datasets. The OAs of the attention-based methods are still higher than those of nonattention-based methods, especially for the Pavia University and Dioni datasets. Compared with other methods, the OAs of the CentralFormer model reach almost the top. It reveals that the ability of the proposal to discriminate features is better even with limited samples. Therefore, the optimal training proportions for four datasets are set to 5%, 2%, 5%, and 5%, respectively.

IV. CONCLUSION

In this article, a CentralFormer model that takes into full account the characteristics of the HSI cube is proposed to represent the relevant attention maps and discriminating spectral–spatial features for HSI classification. It mainly contains the central encoder, the ARE module, and the CERF module. As the core component, the central encoder is composed of an SSLR module, a CMHA module, an FFP, and a CP layer. The SSLR module is exploited to extract the local features. The CMHA module can capture critical attention patterns and effective global features with the help of the relevance map recognized by the ARE module based on both spectral and spatial correlations. After fusing local and global features, a novel CP layer, tailored for HSI classification, remains the relevant characteristic during the pooling procedure of the circles and bands, which not only reduces the computational complexity but also enhances the effectiveness of subsequent attention deduction and feature extraction. Moreover, the CERF module is built to expand the positive effects of the previous relevance map to deeper positions of the network, thereby relieving the deviation of relevance areas. With these elaborate components, the proposed model can extract the discriminating local and global features efficiently for classification. Experimental results on four public HSI datasets demonstrate the structural rationality and outstanding classification performances of the CentralFormer model.

Future studies will mainly be dedicated to enhancing the robustness of the proposed model with limited samples. Improving the attention deduction further for the boundary pixels is a possible direction as well.

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